

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        | (a) |       |  | <p>Windows at 15%                      4 correct                      <b>3 marks</b></p> <p>Floor        at 10%                      2 or 3 correct                      <b>2 marks</b></p> <p>Roof        at 25%                      1 correct                      <b>1 mark</b></p> <p>Walls        at 35%                      0 correct                      <b>0 mark</b></p>                                                                                                                                                                                                                                                                                          |                 | 3   |     | 3     | 1     |      |
|          | (b) | (i)   |  | <p>Fibre-glass is an insulator <b>or</b> it contains air <b>or</b> it contains air pockets <b>or</b> <u>reduces</u> conduction through the ceiling (1) <b>Don't accept</b> traps air</p> <p><u>Reduces</u> convection <u>currents</u> in the loft <b>or</b> air in the loft heats up less (1)</p>                                                                                                                                                                                                                                                                                                                                                                             | 2               |     |     | 2     |       |      |
|          |     | (ii)  |  | <p><b>3 × (1):</b> Ticks in boxes alongside statements 2, 4 and 5 i.e.</p> <p>The required thickness of loft insulation in 2000 is 8 times thicker than in 1970.</p> <p>A house built in 1980 needs 210 mm of loft insulation added to bring it up to 2015 standards.</p> <p>The general trend of the graph indicates that the thickness of required loft insulation has increased at an increasing rate.</p> <p>Deduct 1 mark for each additional tick</p>                                                                                                                                                                                                                   |                 | 3   |     | 3     | 3     |      |
|          |     | (iii) |  | <p><math>4.50 \times 120 = £540</math> (1)</p> <p><math>\frac{540}{98} = 5.51</math> or 5.5 or 6 [years] (1)</p> <p><b>Don't accept</b> 5 [years]</p> <p>Answer of 0.0459 [years] award 1 mark only</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 | 2   |     | 2     | 2     |      |
|          |     | (iv)  |  | <p><b>(1) for either calculation:</b></p> <p>Insulation 1 saving <math>(40 - 5.0) \times £84 = £2\,940</math></p> <p>Insulation 2 saving <math>(40 - 6.0) \times £111 = £3\,774</math></p> <p><b>(1) for second calculation and comment/agreement with builder</b></p> <p><b>Alternative 1:</b></p> <p><b>(1) for either calculation:</b></p> <p><math>5 \times 84 = £420</math> <b>and</b> <math>40 \times 84 = £3\,360</math> <b>and</b> the difference = £2 940</p> <p><math>6 \times 111 = £666</math> <b>and</b> <math>40 \times 111 = £4\,440</math> <b>and</b> the difference = £3 774</p> <p><b>(1) for second calculation and comment/agreement with builder</b></p> |                 |     | 2   | 2     | 1     |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks Available |          |          |           |          |          |
|----------|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  |  |  | <p><b>Alternative 2:</b><br/><math>5 \times 84 = \text{£}420</math> <b>and</b> <math>6 \times 111 = \text{£}666</math> <b>or</b> <math>40 \times 84 = \text{£}3\,360</math> <b>and</b> <math>40 \times 111 = \text{£}4\,440</math> (1)<br/>The differences are <math>\text{£}3\,774</math> <b>and</b> <math>\text{£}2\,940</math> so the builder is correct (1)</p> <p><b>Alternative 3:</b><br/><b>(1) for either calculation:</b><br/><math>3.5 \times 120 = \text{£}420</math> <b>and</b> <math>40 \times 84 = \text{£}3\,360</math> <b>and</b> the difference = <math>\text{£}2\,940</math><br/><math>5.5 \times 120 = \text{£}666</math> <b>and</b> <math>40 \times 111 = \text{£}4\,440</math> <b>and</b> the difference = <math>\text{£}3\,774</math><br/><b>(1) for second calculation and comment/agreement with builder</b></p> <p><b>Alternative 4:</b><br/><math>3.5 \times 120 = \text{£}420</math> <b>and</b> <math>5.5 \times 120 = \text{£}666</math> <b>or</b> <math>40 \times 84 = \text{£}3\,360</math> <b>and</b> <math>40 \times 111 = \text{£}4\,440</math> (1)<br/>The differences are <math>\text{£}3\,774</math> <b>and</b> <math>\text{£}2\,940</math> so the builder is correct (1)</p> <p><b>Alternative 5:</b><br/><b>(1) for either calculation:</b><br/><math>(111 - 84) \times 40 = \text{£}1\,080</math> <b>or</b> <math>666 - 420 = \text{£}246</math><br/><math>\text{£}1\,080 - \text{£}246 = \text{£}834</math> <b>(1) and comment/agreement with builder</b></p> <p>N.B. Any reference to insulation 3 treat as neutral</p> |                 |          |          |           |          |          |
|          |  |  |  | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>2</b>        | <b>8</b> | <b>2</b> | <b>12</b> | <b>7</b> | <b>0</b> |